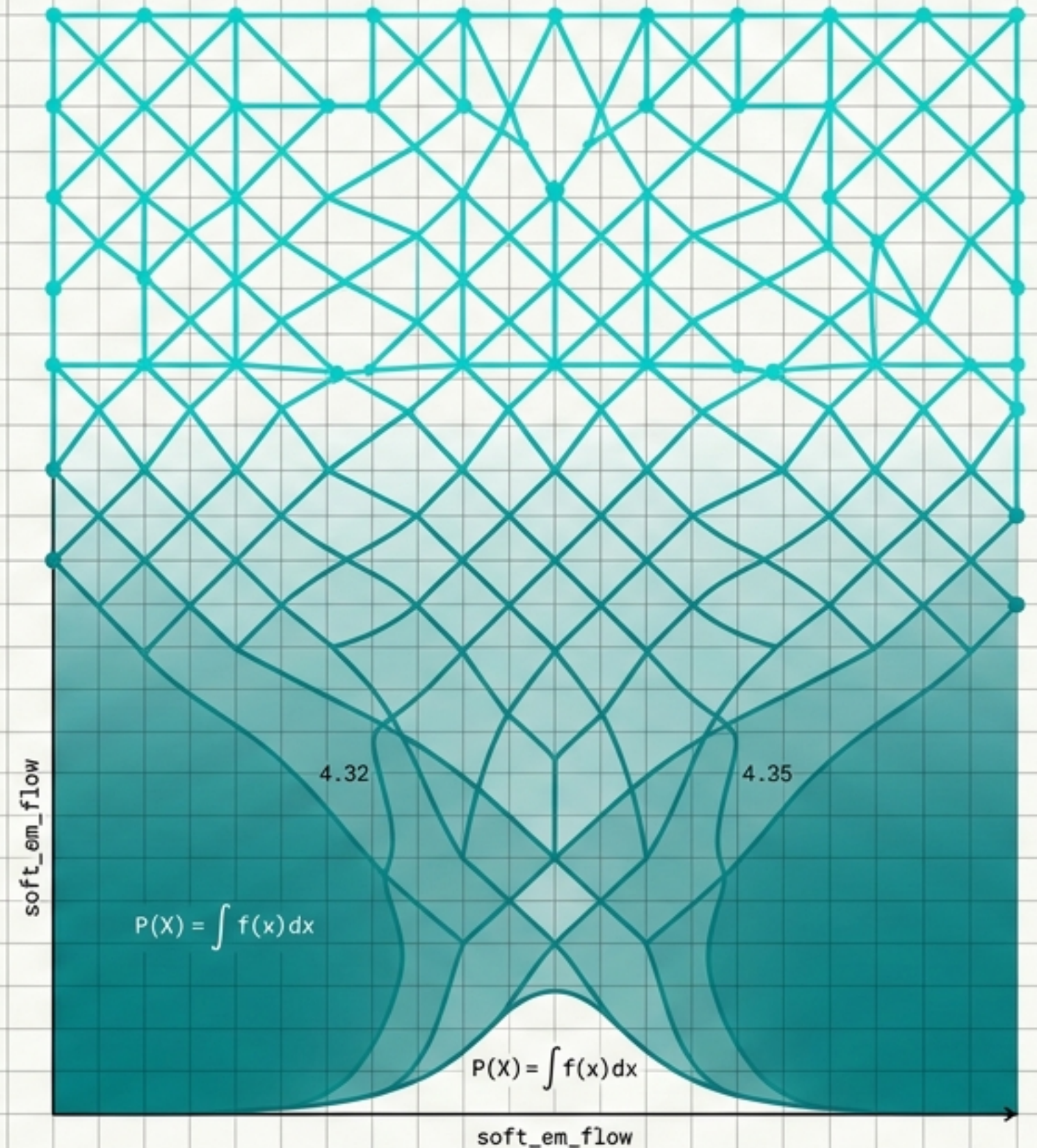


# Portfolio Architecture & The Soft-EM Evolution

Stress-Testing 11 Strategies,  
Eliminating Blind Spots,  
and Validating `gp_cvar_soft`



# The 11-Strategy Taxonomy Matrix

## Baselines

(Validated Reference Models)

**Equal Weight:**  $w_i = 1/N$ .  
No estimation, naive baseline.

**cuFolio:** Max-Sharpe with 3-hop Neumann contagion.

**MPDOK:** Exact-resolvent min-variance + systemic penalty.

## Switchers

(Threshold Logic)

**Smart1 / Smart2:** Hard threshold switching between cuFolio (growth) and MPDOK (defense).

## GP-Posteriors

(The New Frontier)

**gp\_cvar:** Bayesian posterior + Mean-CVaR LP (Hard cut-off).

**gp\_cvar\_soft:** Soft-EM regime mixture.

**gp\_smart\_soft / gp\_smart2\_soft:**  
Switcher pairs incorporating soft-EM.

# Baseline Diagnostics: Deconstructing cuFolio & MPDOK

## cuFolio

$$\mathbf{w} - \underbrace{\gamma\mu}_{\text{Return-Chasing Term}} = (1 + \lambda s) \min \underbrace{\mathbf{w}^T \tilde{\Sigma} \mathbf{w}}_{\text{3-Hop Horizon}}$$

**Return-Chasing Term:** Pulls capital toward recent winners (liability near turning points).

**3-Hop Horizon:** Uses  $k=3$  Neumann truncation. Fails to see deeper systemic contagion.

## MPDOK

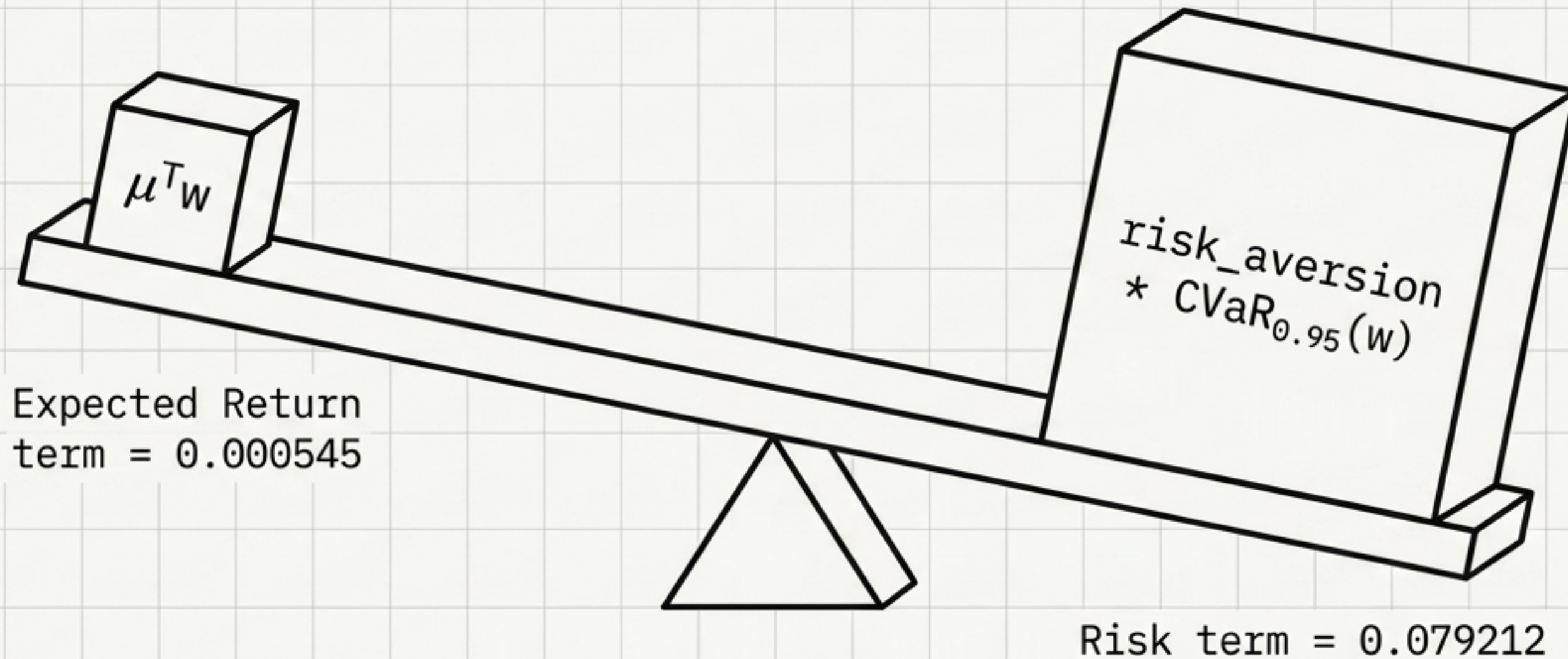
$$\underbrace{(\Sigma + \lambda R)}_{\text{Systemic Penalty}} \mathbf{w} = \frac{0.5}{N} \mid R = \text{diag}(\underbrace{(I - \alpha \hat{A})^{-1}}_{\text{Infinite-Hop Exact Resolvent}})$$

**Systemic Penalty:** Upside Lag. Structurally flows capital to defensives, causing severe drag in trending bull markets.

**Infinite-Hop Exact Resolvent:** Captures total network contagion.

# The GP-Posterior Premise & The 145x Bug

The advanced covariance model barely influenced the weights.



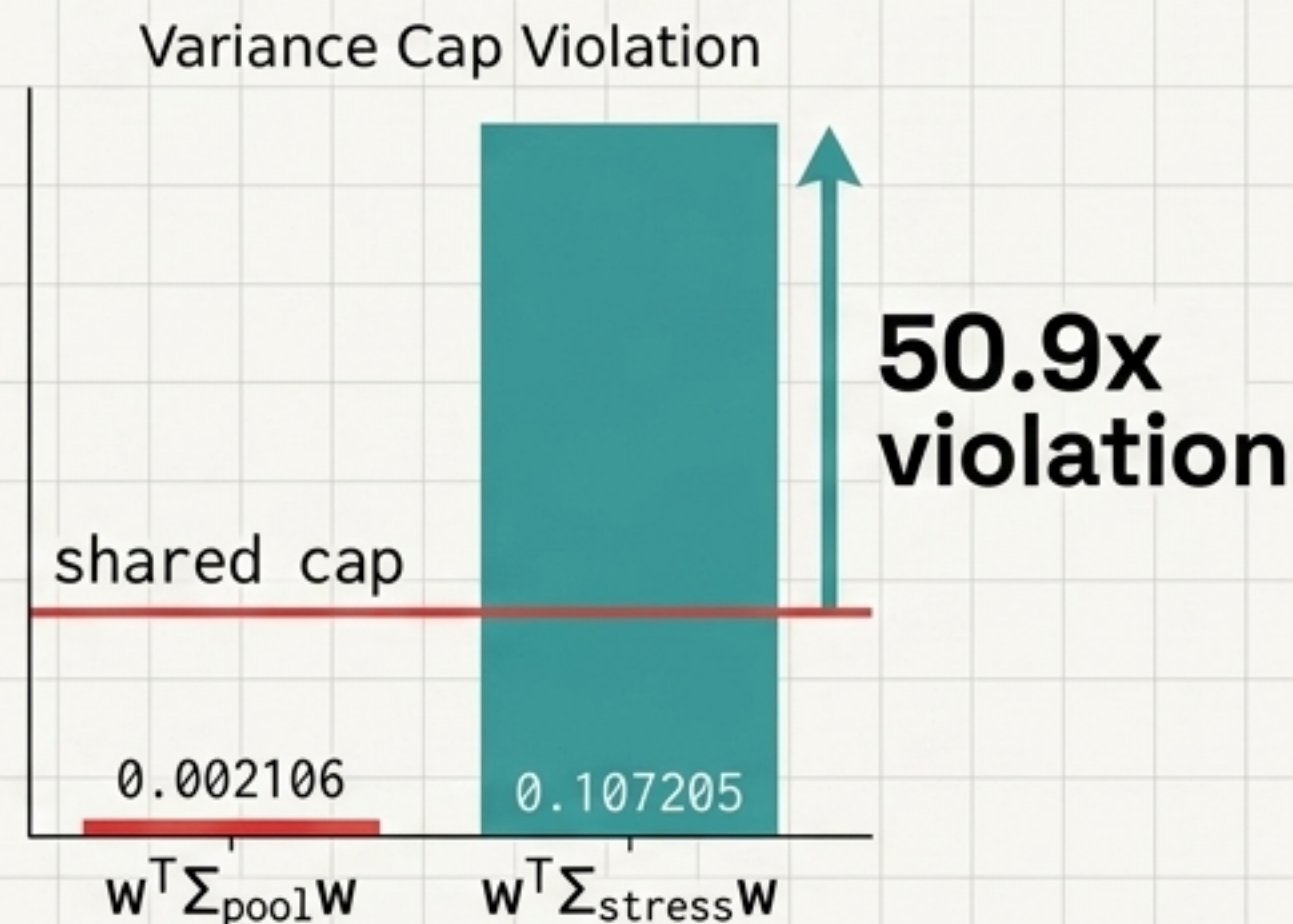
**The Diagnostic:** At the original  $\text{risk\_aversion} = 5.0$ , the objective was 145.5x risk-dominated. The system functioned as a near-pure minimum-CVaR allocator.

# The Danger of Hard Cut-Offs in Regime Classification

## Failure 1: The Hard Cut-off Chasm

Real daily return isn't neatly bimodal. The original Gaussian Mixture Model **forced a hard cut**, inadvertently carving out "quiet days vs everything else" instead of "rare crises vs normal." Result: Vital stress data was completely discarded.

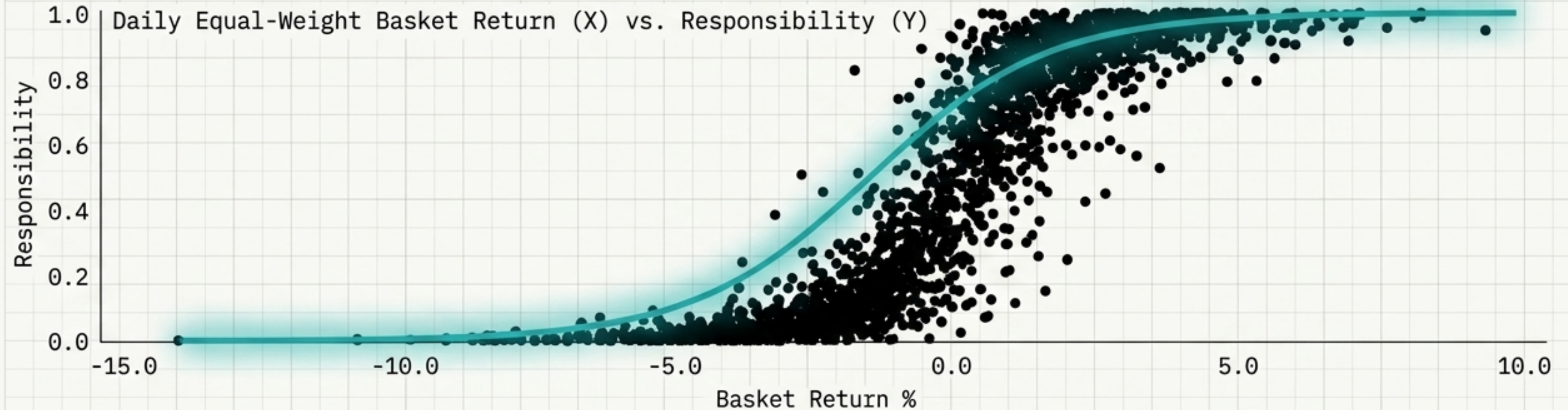
## Failure 2: The Mismatched Variance Cap



Fitting a "stress" regime using a pooled variance cap reference is mathematically fatal. The optimizer blew through the shared cap by 50.9x because real crisis targets naturally possess higher cross-asset dispersion.

# The Soft-EM Solution: Responsibility Weighting

No data discarded. Every single day contributes to both regime fits based on posterior responsibility.



## Downside-Only Filtering

Replaced absolute return with signed return. A massive up-day is no longer classified as "systemic risk".

## Data Proof

The 10 worst basket return days all hit ~0.83 responsibility; the 10 best days map strictly to 0.00.

# Sweeping the Optimizer: Balancing Risk and Return



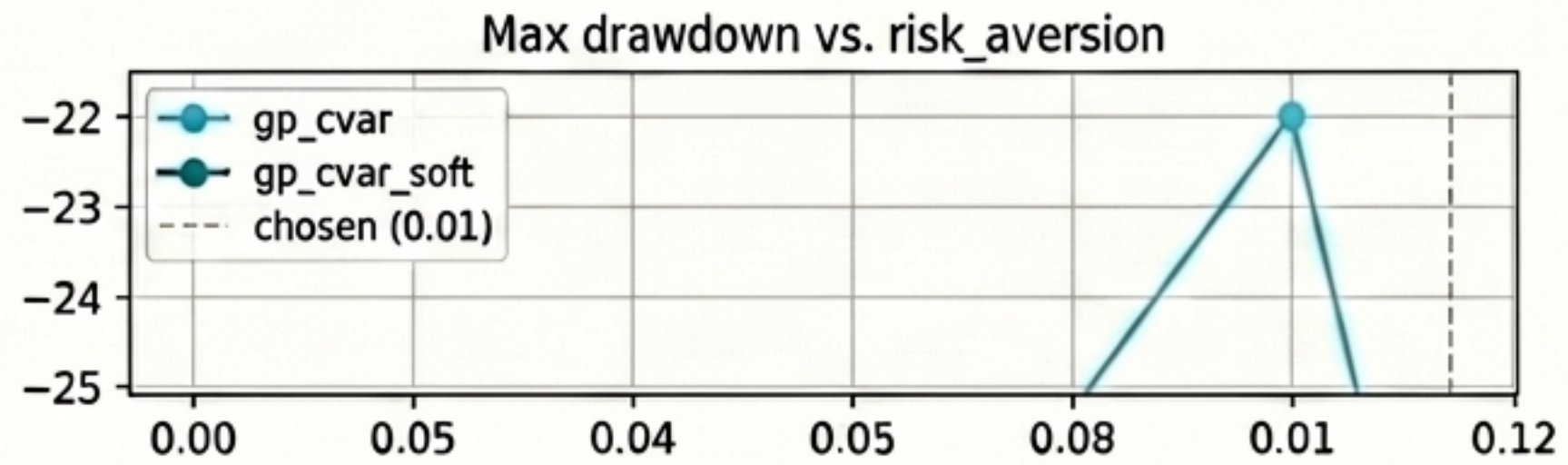
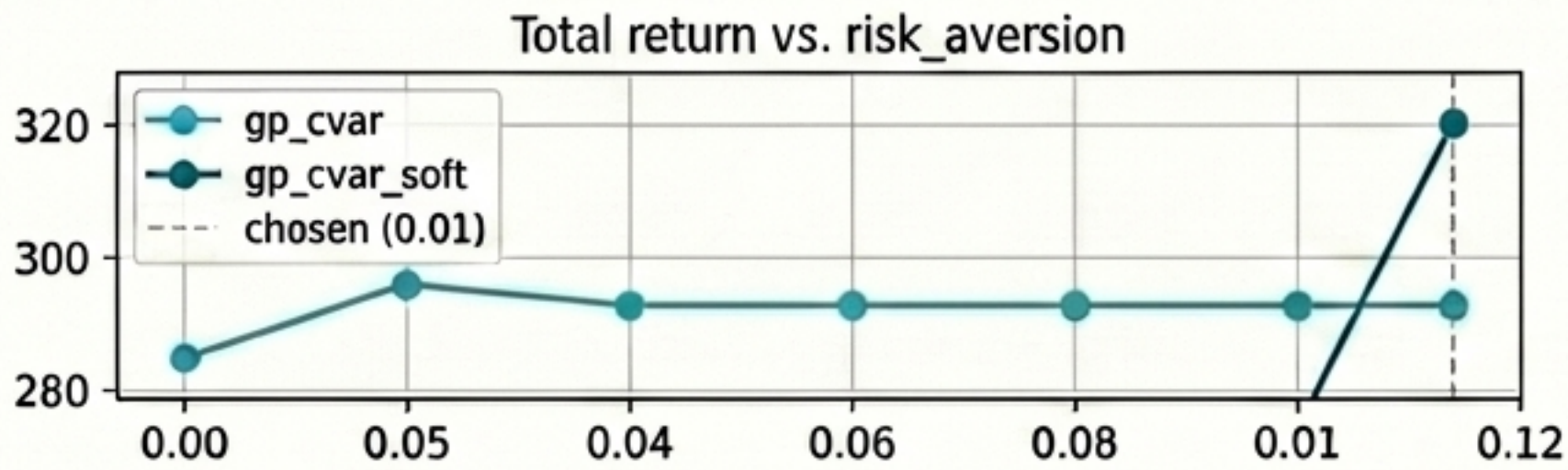
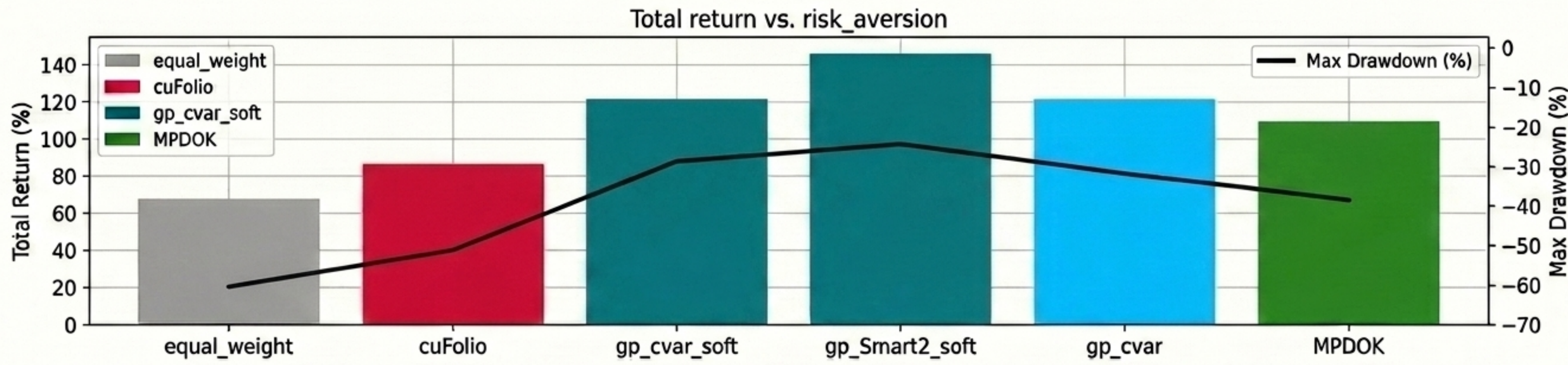
## Analysis Panel

A shared `risk_aversion` parameter was swept from 5.0 down to 0.01 to find the optimal balance.

**The Discontinuity:** LP corner solutions jump discontinuously as box-constraint caps change (non-monotonic structure).

**The Selection (0.01):** Selected because `gp_cvar_soft` asserts **dual-axis** dominance here. It explicitly wins on both metrics simultaneously compared to any hard variant.

# Bull Market Performance (2020-2026)

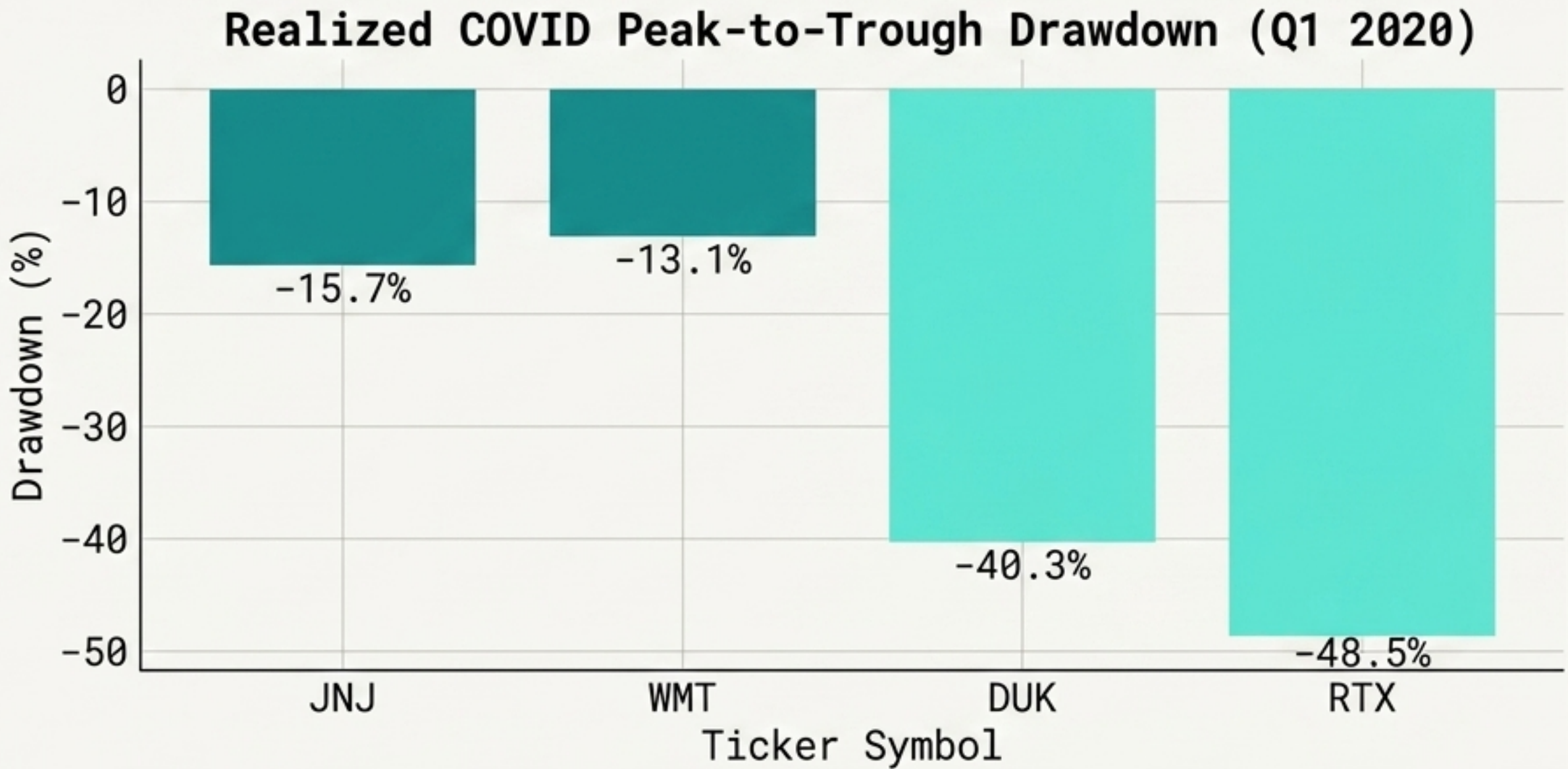


- **The Baseline Win:** `gp_cvar_soft` achieves 320.6% Return / -29.4% Drawdown, crushing equal weight and cuFolio.
- **The Apex Combinator:** `gp_Smart2_soft` is the board leader: 340.9% Return and a resilient -25.2% Max Drawdown.
- **The Soft-EM Edge:** The `soft variants` consistently improve max drawdown over every hard GP variant while capturing more upside.

# Case Study (Section 3.1): Anatomy of a 4-Name Swap

Out of an 8-name portfolio, the models diverged on exactly 2 names prior to the COVID crash.

| The Swap Matrix |                                                                         |
|-----------------|-------------------------------------------------------------------------|
|                 | Hard Model Picked:<br><b>DUK</b> (Utility) &<br><b>RTX</b> (Aerospace)  |
|                 | Soft Model Picked:<br><b>JNJ</b> (Healthcare) &<br><b>WMT</b> (Staples) |



**Result: A massive, directional outperformance shielding the portfolio during the V-shaped panic.**

# Why the Math Picked Quality (The Input-Output Link)

**The Engine:** The optimizer was running near a pure return-maximizing corner; the cross-asset kernel shrinkage mean estimate (`mixed_mean`) did the heavy lifting, not explicit covariance reasoning.

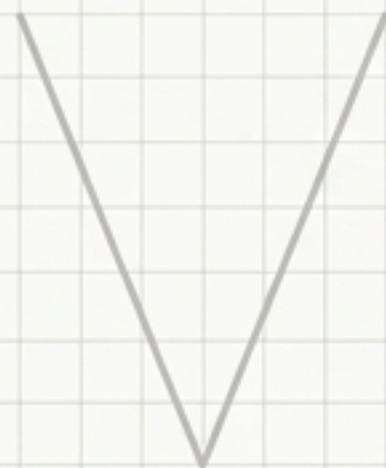
| Ticker | Calm Mean | Stress Mean | Mixed Mean ( <code>mixed_mean</code> ) |
|--------|-----------|-------------|----------------------------------------|
| RTX    | 0.17%     | -1.88%      | 0.062%                                 |
| WMT    | 0.14%     | -1.18%      | 0.074%                                 |

**The Logic:** JNJ and WMT won because their calm-day means remained strong, while their stress-day means were significantly less devastating than RTX's -1.88% daily hit.

**Practitioner Caveat:** This is a documented instance of the mechanism operating rationally in one specific regime. It is economic validation of the math, not proof of a universal market law that will perfectly predict every future rotation.

# Stress-Testing Architectures: Three Historical Crashes

## V-Shaped Panic (COVID 2020)



- Train: 2015-2019 (1257d)
- Test: 2020
- Characteristics: Fast, severe snapback.

## Grinding Bear (2022)



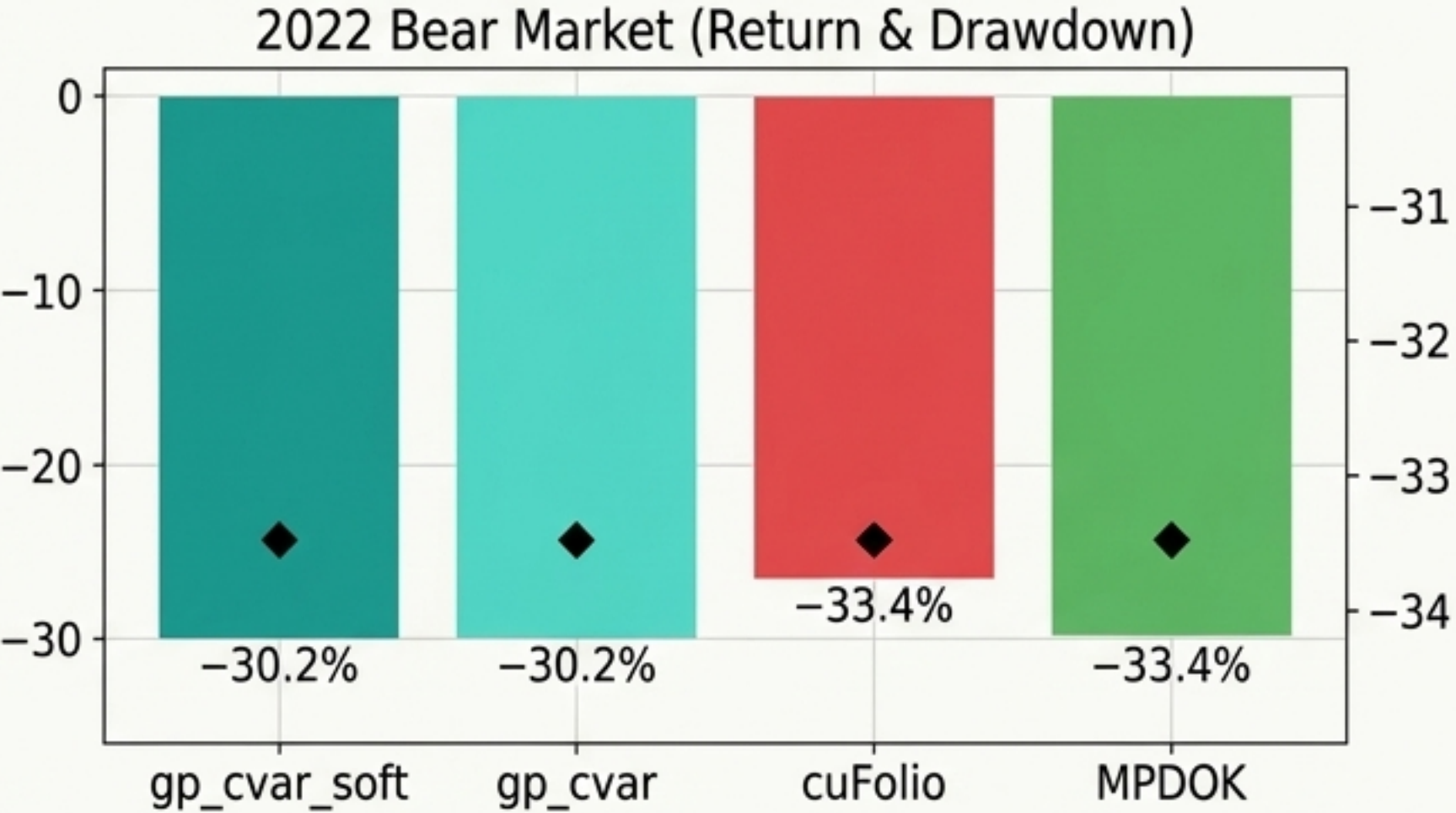
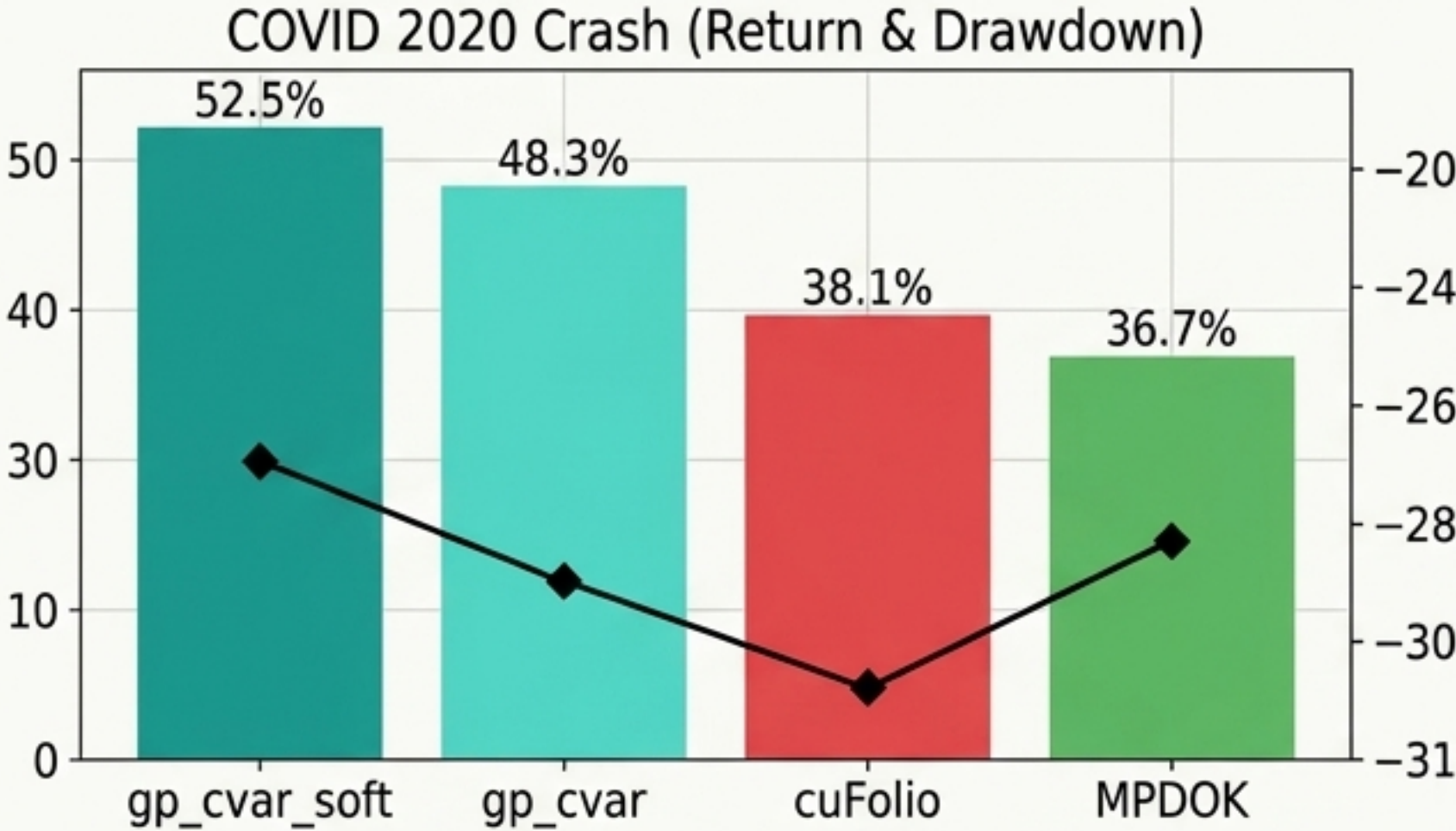
- Train: 2017-2021 (1258d)
- Test: 2022
- Characteristics: Correlation-driven rate-hike environment, no immediate recovery.

## Systemic Collapse (2008 GFC)



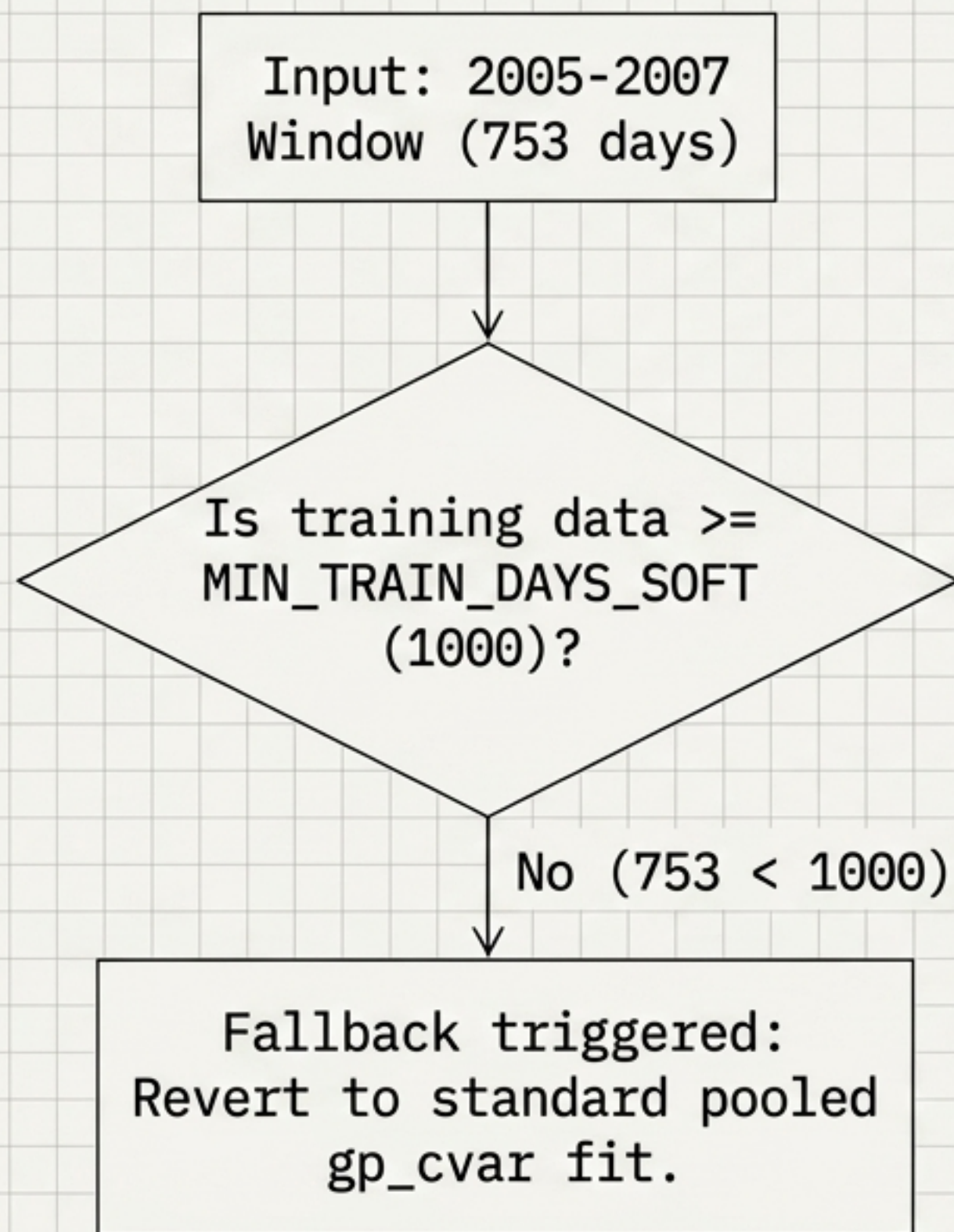
- Train: 2005-2007 (753d)
- Test: 2008-2009
- Characteristics: Deep structural failure, distinctly thin training data.

# Stress Tests 1 & 2: V-Shapes and Grinding Bears



| COVID 2020 Results Analysis                                                                                                                                                              | 2022 Bear Results Analysis                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| gp_cvar_soft outright beats gp_cvar on both axes (Return 52.5% vs 48.3%, Max DD -25.2% vs -30.0%). The core idea is validated in an actual panic.                                        | An exact mathematical tie between hard and soft variants (Return -30.2%, Max DD -33.4%).<br><br>Neither GP variant beats cuFolio or MPDOK in 2022. |
| <b>Analytical Insight:</b> MPDOK's exact closed-form resolvent math is fundamentally better suited to a grinding, high-correlation rate-hike decline than ANY GP-fitted covariance model |                                                                                                                                                    |

# Stress Test 3: The 2008 Thin-Data Guardrail



## The Problem

The 2005-2007 window provides only 753 training days. With a 5% tail cut, that leaves only ~38 effective "stress days" to fit a full GP component.

## The Vulnerability

Without protection, `gp_cvar_soft` actively underperforms due to unreliable regime-splitting on sparse data.

## The Result

Stabilizes the 2008 performance (-3.5% return / -56.5% max dd), preventing algorithmic degeneration. Honesty in design builds trust.

# Final Synthesis: Eliminating the Blind Spots

|                              |                                                                  |                                                                                                    |
|------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>The Contagion Horizon</b> | <b>Blind Spot:</b> cuFolio truncates at 3 hops.                  | <b>Solution:</b> Smart2 integrates MPDOK's infinite-hop exact resolvent. ✓                         |
| <b>The Regime Data Purge</b> | <b>Blind Spot:</b> Hard cut-offs discard vital crisis data.      | <b>Solution:</b> Soft-EM responsibility weighting preserves continuous data contribution. ✓        |
| <b>Objective Imbalance</b>   | <b>Blind Spot:</b> GP model running as a 145x risk-dominated LP. | <b>Solution:</b> Balanced risk_aversion = 0.01 unlocking simultaneous return/drawdown dominance. ✓ |

# Limitations & The Road Ahead

## Constraint 1: The Long-Only Boundary.

The Mean-CVaR LP has  $w_{\min}=0$ . True short-hedging (shorting a specific name solely for its crisis-correlation structure) **cannot be expressed**.

## Constraint 2: Hyperparameter Overfitting.

The optimal `risk_aversion = 0.01` was swept over the reported backtest history. This carries **acknowledged overfitting risk**; it is **not out-of-sample validated yet**.

## Constraint 3: Static Buy-and-Hold Limitations.

The backtests evaluate a static **hold-and-rebalance** approach. The weights do **not dynamically adapt mid-crash** as a live **walk-forward** deployment would.

**Portfolio Studio  
validates the  
architecture.  
Walk-forward live  
adaptation is the  
next engineering  
frontier.**